Marketing Campaign Analysis Report

Facebook vs AdWords Campaign Analysis

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1. Executive Summary

This report analyzes the performance of Facebook and AdWords advertising campaigns using statistical and analytical methods including:

- Descriptive statistics
- A/B testing (T-test)
- Regression analysis
- Time series analysis

Key findings indicate that **Facebook significantly outperforms AdWords in terms of conversions**, and that **clicks are a strong driver of conversions**, making them a critical optimization metric.

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2. Objective

The primary objectives of this analysis are:

- Compare the performance of Facebook and AdWords campaigns
- Determine if differences in performance are statistically significant
- Identify key factors influencing conversions
- Build predictive understanding of campaign performance
- Analyze trends over time

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3. Data Overview

The dataset includes key advertising metrics such as: Ad Views, Clicks, Conversions, Cost, Click Through Rate (CTR), Conversion Rate, Cost Per Click (CPC). These metrics are analyzed separately for Facebook and AdWords.

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4. Exploratory Data Analysis (EDA)

Key Observations:

- **AdWords generates higher impressions**, but **Facebook generates higher conversions**
- Facebook shows: Higher **conversion rate,** better **efficiency per click**
- AdWords: Higher reach but **lower conversion efficiency**

	Facebook		Adwords	
	Mean	**Median**	**Mean**	**Median**
Ad Views	2179.69	2202.00	4717.20	4711.00
Clicks	44.05	43.00	60.38	60.00
Conversions	11.74	12.00	5.98	6.00
Cost	87.78	88.00	134.98	135.00
CTR	2.20	2.00	1.30	1.29
Conversion Rate	27.15	26.77	10.18	9.93
CPC	2.19	2.02	2.38	2.29

Insight: Higher visibility (AdWords) does not necessarily translate into higher conversions. **Facebook is more effective in converting users.**

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5. A/B Testing (T-Test Results)

Objective: To determine whether the difference in conversions between Facebook and AdWords is statistically significant.

Results: Facebook Mean Conversions: **11.74.** AdWords Mean Conversions: **5.98.** p-value: **~0 (extremely small).**

Interpretation: Since **p-value < 0.05**, we reject the null hypothesis. There is a **statistically significant difference** between the two platforms

Insight: Facebook consistently delivers **higher conversions**, not due to randomness but due to actual performance differences.

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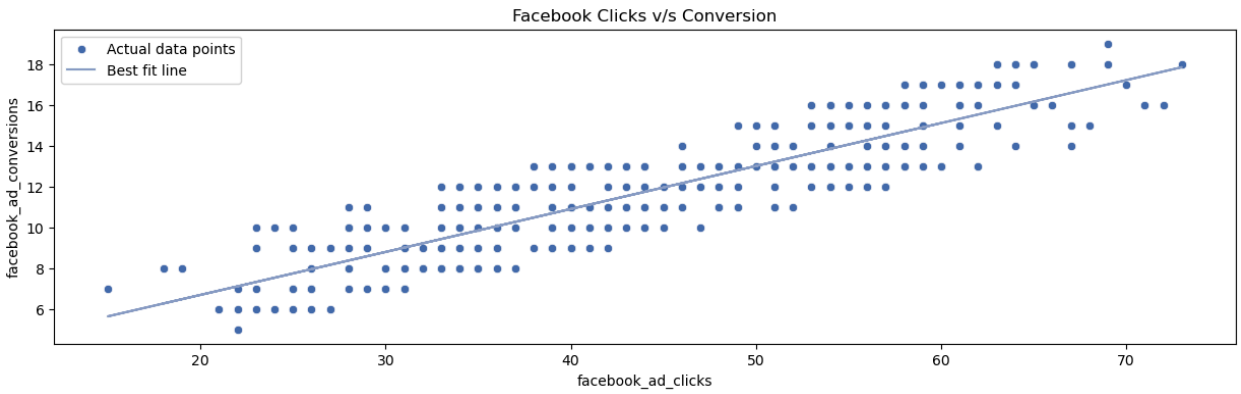
6. Regression Analysis (Facebook Performance)

Objective: To understand how clicks influence conversions.

Model Performance: R^2 Score: **76.35%,** Mean Squared Error: **2.02**

Interpretation: ~76% of conversion variability is explained by clicks. Prediction error is low (2.02), indicating a **reliable model**

Key Findings: Strong **positive relationship** between clicks and conversions.



Insight: Clicks are a **primary driver of conversions**, making them a key optimization lever.

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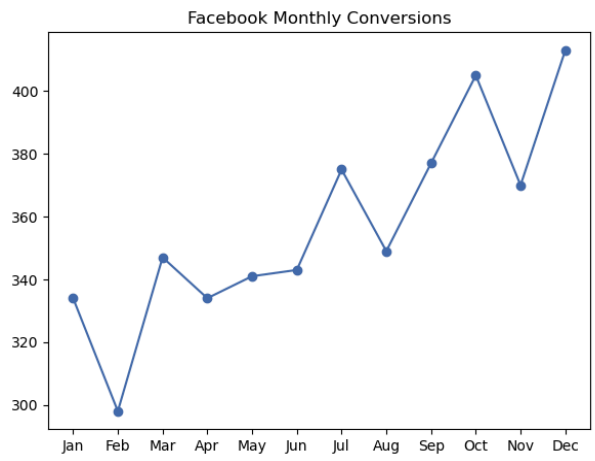
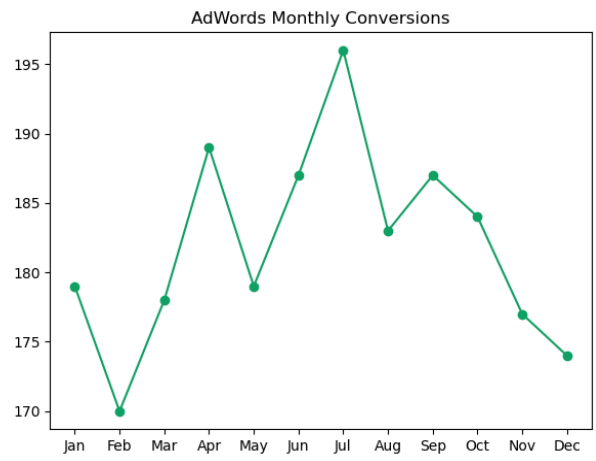
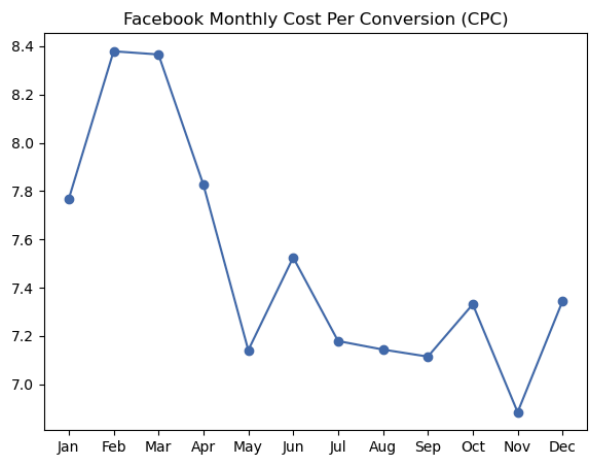
7. Time Series Analysis

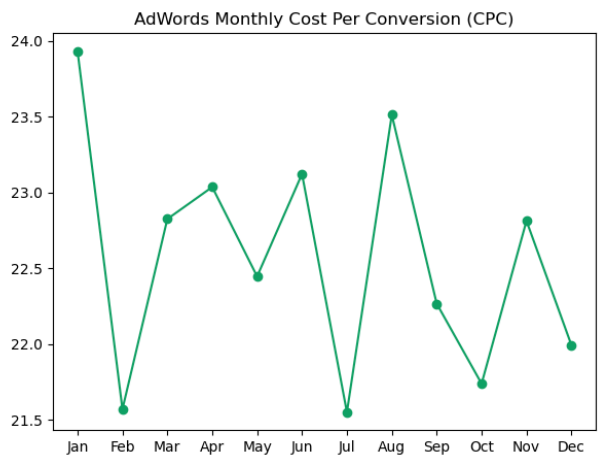
Objective: To analyze trends and patterns over time.

Key Observations: Certain periods exhibit **higher engagement and conversions.** Possible presence of Seasonal trends or Campaign effectiveness cycles.









Insight: Timing plays a role in campaign success. Optimizing campaigns based on **high-performing periods** can improve outcomes.

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8. Key Insights Summary

- Facebook is **significantly more effective** than AdWords in generating conversions
- Higher impressions do not guarantee better performance
- Clicks are a **strong predictor of conversions**
- Campaign performance varies over time and should be monitored dynamically
- Statistical testing confirms results are **reliable and not random**

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9. Business Recommendations

1. Budget Allocation: Allocate more budget to **Facebook campaigns.** Reduce spending on underperforming AdWords segments.

2. Optimization Focus: Focus on increasing **clicks** to improve conversions. Improve ad creatives and targeting to boost CTR.

3. Performance Monitoring: Track campaigns over time to identify **high-performing periods.** Adjust campaigns dynamically based on trends.

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10. Conclusion

This analysis demonstrates that:

- **Facebook is the superior platform** for conversions in this dataset
- Statistical and predictive methods confirm the reliability of findings
- Data-driven optimization can significantly improve marketing performance